

GAJENDRA VERMA

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TECHNICAL SKILLS

Languages: JavaScript, Python, C++, C

Backend: Node.js, Express.js, REST APIs, Server-Sent Events (SSE), JWT, Clerk

Frontend: React.js, Redux Toolkit, Tailwind CSS, HTML, CSS

Databases & Storage: MongoDB, ChromaDB, MySQL, SQL, Cloudinary

AI / ML: RAG, Transformers.js, Mistral AI, Scikit-learn, NumPy, Pandas, Matplotlib

Developer Tools: Git, GitHub, Postman, VS Code, Google Colab, Jupyter Notebook

Core CS: DSA, OOP, DBMS, Operating Systems, System Design (HLD)

PROJECTS

Konnect - Full Stack Social Media Platform (🔗 [GitHub](#) | 🌐 [Live Demo](#))

Tech Stack: React, Redux Toolkit, Tailwind CSS | Node.js, Express.js, MongoDB | Clerk, Inngest

- Built a full-stack social platform (feed, posts, stories, connections, chats) using **MERN stack**, with **Clerk**-based authentication, webhook-driven user synchronization, and **Redux Toolkit** for centralized state management.
- Implemented real-time messaging using **Server-Sent Events** over standard HTTP, maintaining an in-memory registry mapping user IDs to response streams to push message, edit, and delete events instantly, with automatic reconnection handled by the browser's native EventSource API.
- Encrypted chat messages using **AES-256-CBC** with a random IV per message, preventing identical plaintexts from producing identical ciphertext.
- Designed **30 authenticated REST API endpoints** across posts, users, messaging, and stories, protected by **Clerk** auth middleware, with MongoDB schema modeling a hybrid social graph and **24-hour** rate limiting on connection requests.

StudyStream - AI-Powered Lecture RAG Platform (🔗 [GitHub](#) | 🌐 [Live Demo](#))

Tech Stack: Node.js, Express.js, React | ChromaDB, Transformers.js | Mistral AI

- Built a **Node.js-based RAG (Retrieval-Augmented Generation)** platform that converts YouTube lectures into searchable knowledge by extracting timestamped captions, chunking transcripts, generating local embeddings with **Transformers.js**, and indexing them in **ChromaDB**.
- Implemented **semantic question answering** using top-K vector retrieval and **Mistral AI**, returning transcript-grounded answers with clickable YouTube timestamp citations derived directly from retrieved chunk metadata.
- Developed an interactive React dashboard with **real-time SSE processing updates**, AI-generated summaries and key concepts, an auto-generated quiz, and an AI Tutor chat grounded in the transcript.
- Built a custom **Recall@K retrieval evaluation** against a 30-question ground-truth benchmark, measuring **70.00% Recall@5** (46.67% Recall@1, 66.67% Recall@3) independently of answer generation.

FORENSIGHT - Criminal Face Recognition System (🔗 [GitHub](#) | 🌐 [Live Demo](#))

Tech Stack: React, CSS Modules, Vite | Node.js, Express.js, MongoDB | Python, DeepFace, FaceNet, JWT, Cloudinary

- Built a criminal identification system with role-based (**NORMAL/ADMIN**) **JWT** authentication and **MongoDB**-based suspect record management, supporting profile enrollment with age, sex, and crime-history details, with photos stored via **Cloudinary**.
- Developed an **HTML5 Canvas** facial-composite editor with priority-based layer rendering, drag/resize/rotate interactions, and z-order controls to construct a suspect's face from modular parts.
- Spawned a Python subprocess running **DeepFace (FaceNet model)** to generate **128-dimensional**, L2-normalized face embeddings from the composite image, then ranked candidates by **cosine similarity** against MongoDB-stored embeddings to return the **top 5** matches, computed in-process without a dedicated **vector database**.

EDUCATION

Maulana Azad National Institute of Technology, Bhopal

2023 – 2027

B.Tech in Electronics and Communication Engineering | CGPA: 8.99

Rani Laxmi Bai Memorial School, Lucknow

2022

Class XII | CBSE | 95.8%

Rani Laxmi Bai Memorial School, Lucknow

2020

Class X | CBSE | 91.8%

POSITIONS OF RESPONSIBILITY

Management Head, Roobaroo Cultural Society, MANIT Bhopal

Jan 2026 – Present

- Coordinate rehearsals and mentor junior performers across vocals and instruments for Roobaroo's program at Maffick/Techno's Gig-a-Night, performing before crowds of **1,000+**.

Student Mentor, MANIT Bhopal

Aug 2025 – Present

- Mentor **6 first-year students** through academic guidance and onboarding support.

ACHIEVEMENTS

- Solved **500+** **DSA** problems across LeetCode, GeeksforGeeks, CodeStudio, and other coding platforms.
- Selected among the **Top 50 teams** from MANIT Bhopal for the **Smart India Hackathon (SIH) 2025**.
- Achieved a **1900 (Rapid)** rating on Chess.com, placing in the **top 1.5%** of rated players.